Fall 2026

Technical and Academic English
for Computer Science

Writing tool, LaTeX

Lecture #2

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Content

Content

- **LaTeX Introduction**
- LaTeX Basic
- Special characters
- Layout
- Custom commands
- Lists
- Images
- Tables
- Mathematics
- Listings
- Bibliography
- Templates

LaTeX

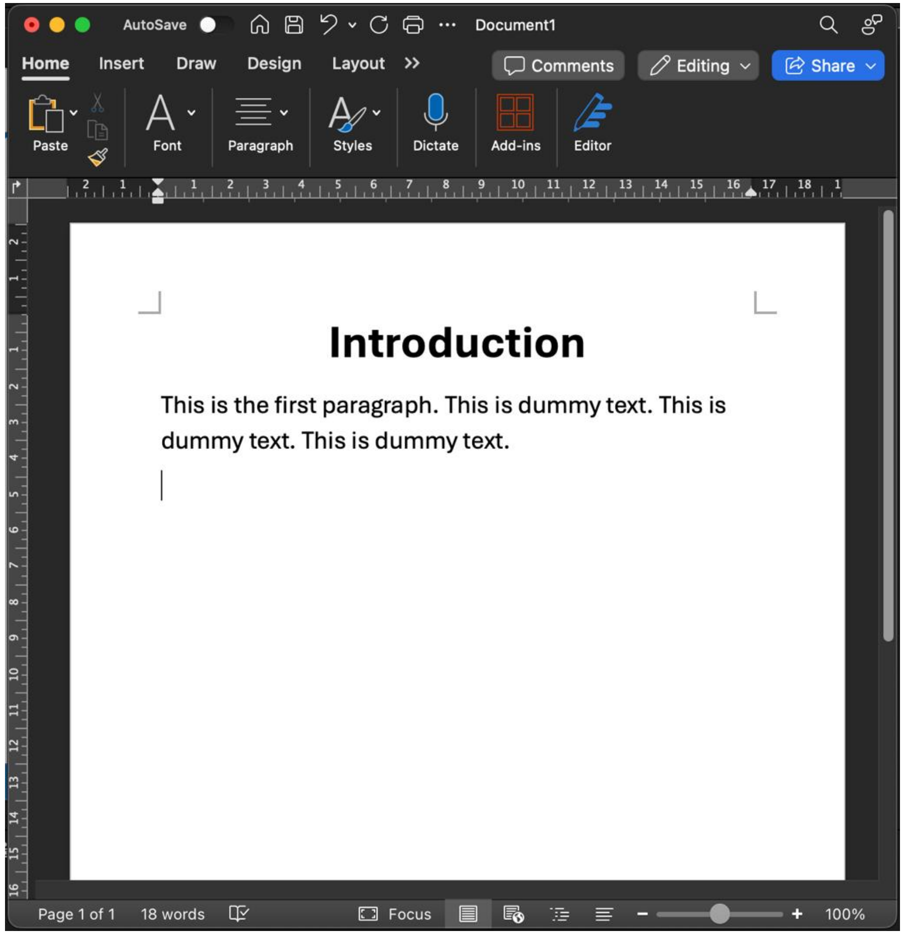
- LaTeX (pronounced either "Lah-tech" or "Lay-tech") is a software for typesetting documents.
- It's a document preparation system.
- It is not a word processor, but is used as a document markup language.
- It is a free, open source software.
- It was originally written by *Leslie Lamport* and is based on the TeX typesetting engine by *Donald Knuth*.

LaTeX and Word

<LaTeX>

```latex
\begin{document}

\maketitle

\section{Introduction}
This is the first
paragraph. This is dummy
text. This is dummy text.
This is dummy text.

\end{document}
```

<Word>



LaTeX: Benefits

- Well-suited for scientific and technical documents.
- Superior for typesetting of mathematical formulas.
- High-quality, stable output, handles complex documents easily, no matter how large they are.
- Built-in automation for cross-referencing, numbering, and the generation of lists of contents, figures, tables, indexes, glossaries, and bibliographies.

LaTeX: Benefits

> Multilingual support with language-specific features.

> Various output formats: PostScript, PDF, DVI, and HTML.

> Template flexibility: letters, presentations, bills, philosophy books, law texts, music scores, and even for chess game notations.

> Cross-platform consistency:
>
> - Same output on Windows, Linux, Mac OS X, and others

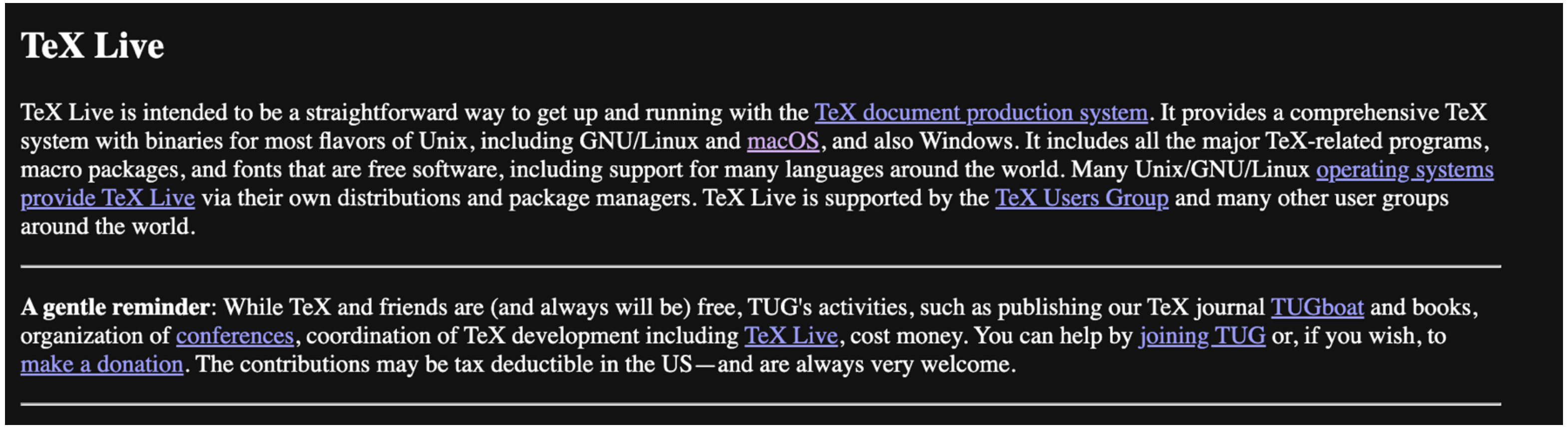
> Plain text workflow – readable and editable.

LaTeX: Disadvantages

- Long learning curve
- Difficulty of understanding large documents
- Challenging to create and design tables
- Expensive troubleshooting/debugging effort

LaTeX Distributions

> **Tex Distributions**
- Tex Live
- MikTeX



TeX Live

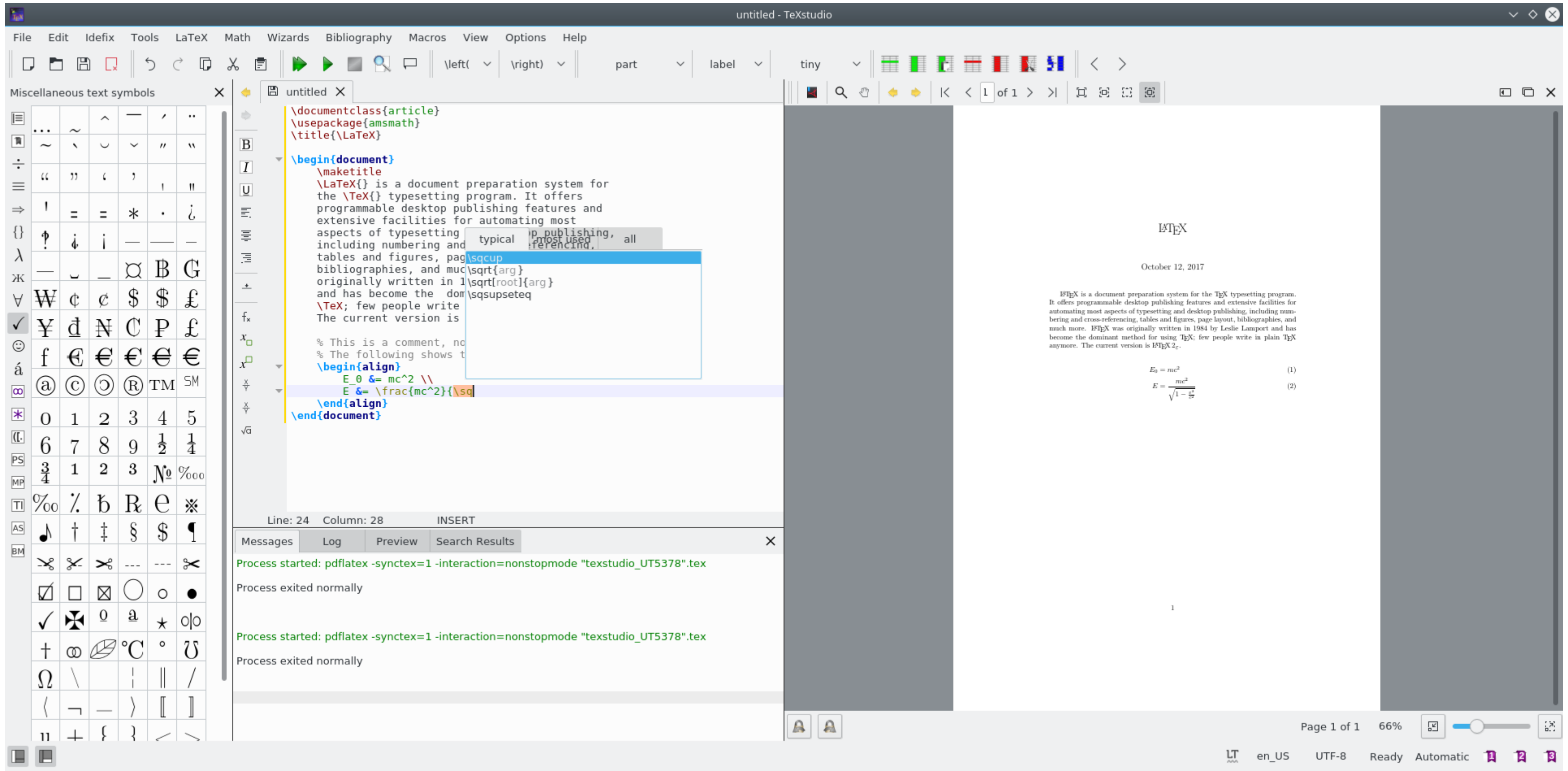
TeX Live is intended to be a straightforward way to get up and running with the TeX document production system. It provides a comprehensive TeX system with binaries for most flavors of Unix, including GNU/Linux and macOS, and also Windows. It includes all the major TeX-related programs, macro packages, and fonts that are free software, including support for many languages around the world. Many Unix/GNU/Linux operating systems provide TeX Live via their own distributions and package managers. TeX Live is supported by the TeX Users Group and many other user groups around the world.

A gentle reminder: While TeX and friends are (and always will be) free, TUG's activities, such as publishing our TeX journal TUGboat and books, organization of conferences, coordination of TeX development including TeX Live, cost money. You can help by joining TUG or, if you wish, to make a donation. The contributions may be tax deductible in the US—and are always very welcome.

> **Provides**
- Latex engines:
  - pdfLaTeX — Traditional, mature, and highly compatible
  - XeLaTeX — Native Unicode and system font support
  - LuaLaTeX — Unicode/font support + Lua programmability
- LaTeX packages
- Fonts
- Supporting tools

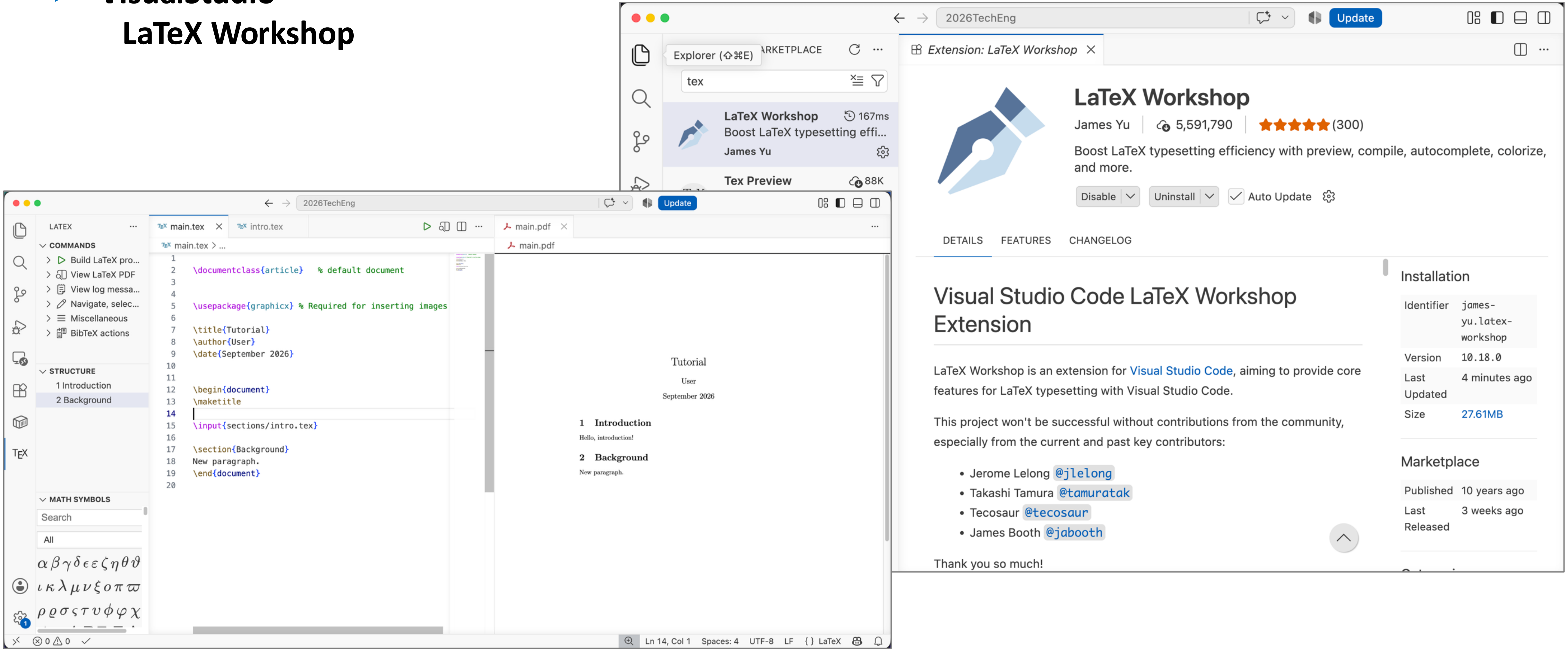
LaTeX: Editors

> **Editors**
- TeXstudio
- TeXworks
- TexMaker
- TeXnicCenter
- Gummi
- Lyx



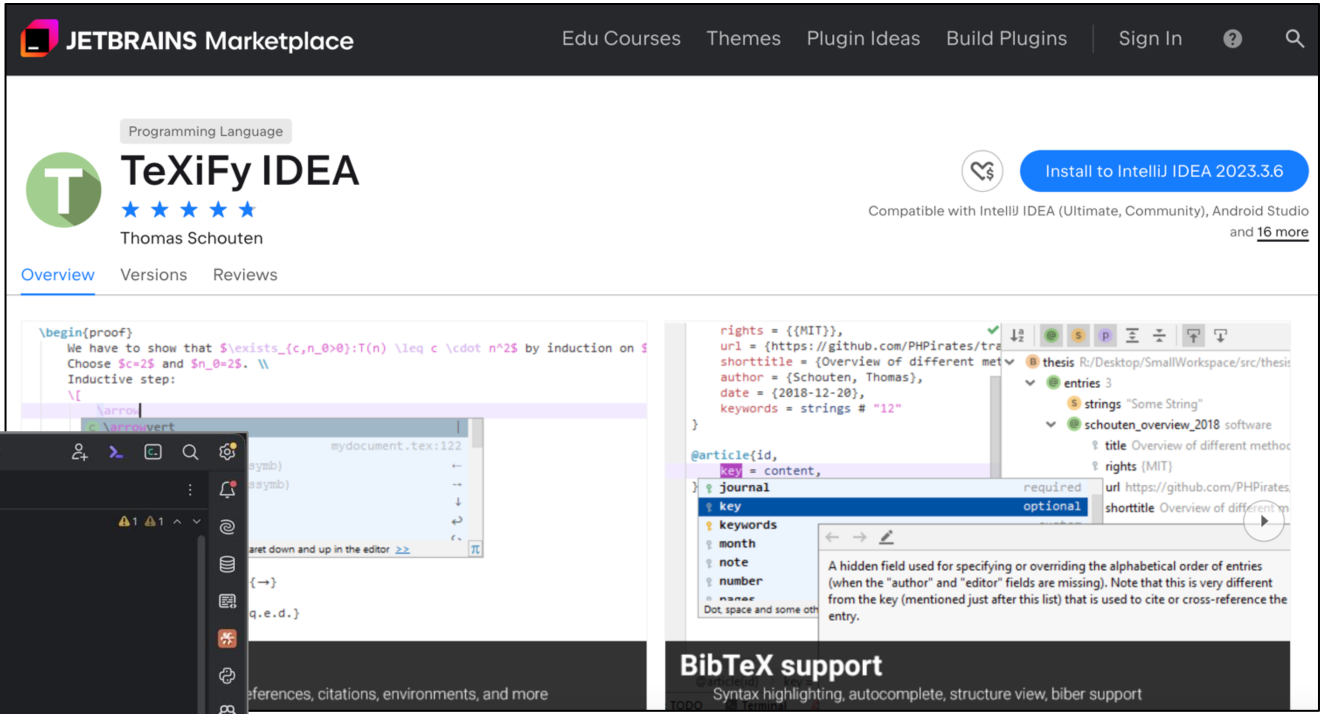
LaTeX: Editors (Plugins)

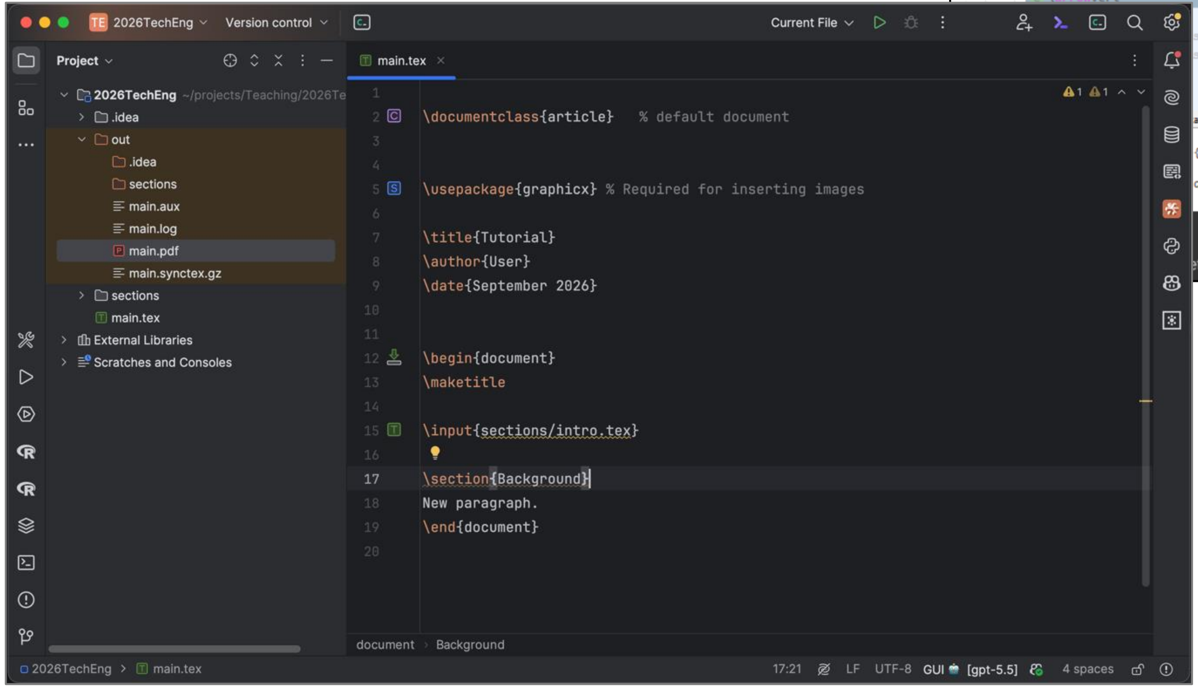
> **VisualStudio – LaTeX Workshop**



LaTeX: Editors (Offline-Plugins)

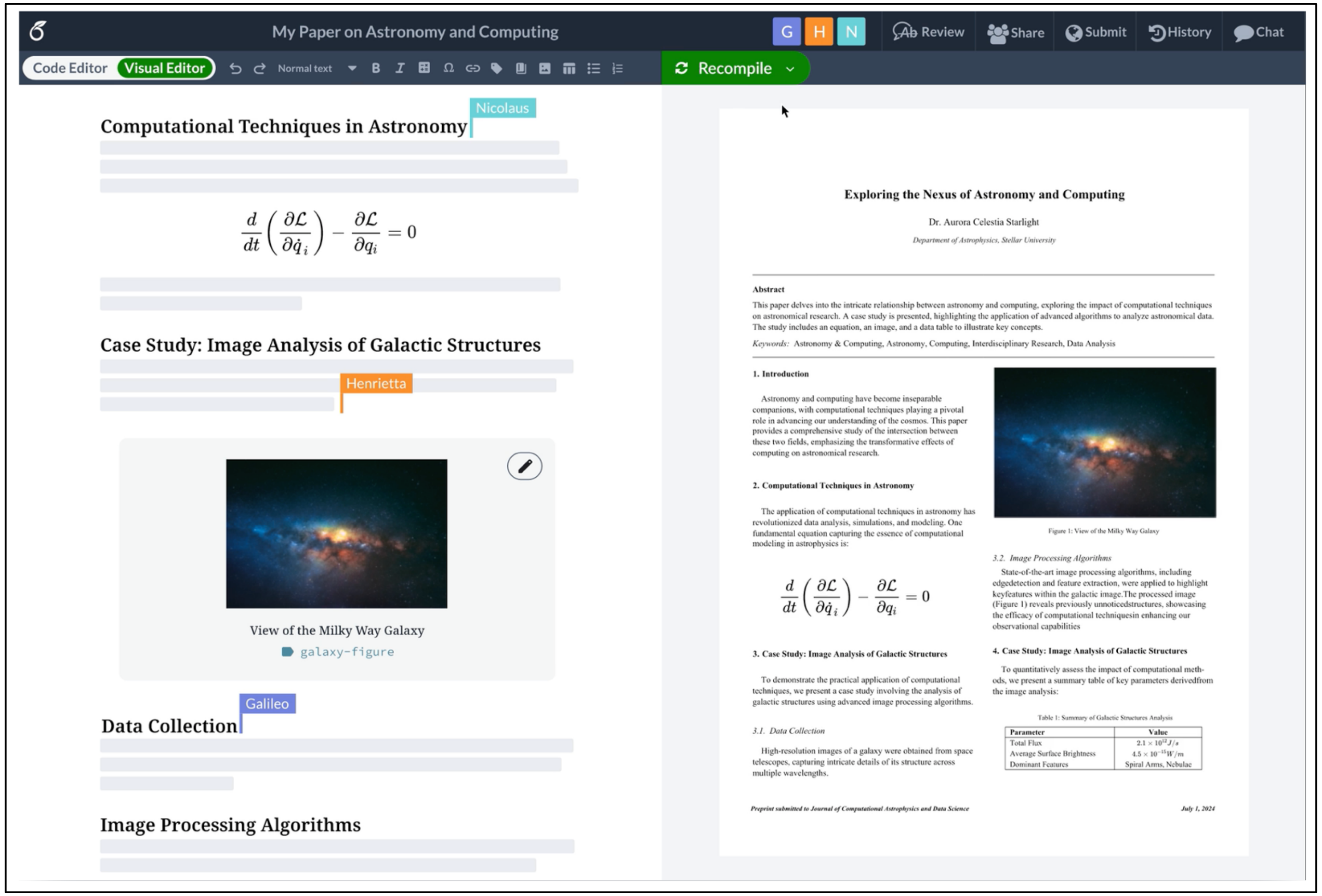
> **IntelliJ IDEA**
- TeXiFy IDEA
- PDF Viewer





LaTeX: Editors (Online)

- **Overleaf**
- Papeeria
- CoCalc
- Authorea



Content

A Typical LaTeX Document

Preamble

```latex
\documentclass[a4paper, 12pt]{article}

% Packages
\usepackage{amsmath}
\usepackage{graphicx} % Required for inserting images

% Variables
\title{Introduction to LaTeX}
\author{Author Name}
\date{\today}
```

Command

Body

```latex
% Document Body
\begin{document}
\maketitle
Hello, world!
\end{document}
```

A Typical LaTeX Document (Output)

Introduction to LaTeX

Author Name

September 5, 2026

Hello, world!

LaTeX Commands

- **LaTeX commands** begin with **a backslash(\)**, followed by big or small letters.
- Commands may have arguments, given in curly braces or in square brackets.
- Calling a command looks like the following:

```
\command[optional argument]{argument}
```

- There could be several arguments, each of them in braces or brackets.

```
\command[optional1][optional2]{argument1}{argument2}{argument3}
```

- **Arguments in curly braces are mandatory.**

Spaces, Line Breaks, and Paragraphs

```latex
\begin{document}

\maketitle

\section{Introduction}
This                  is the
first paragraph. This is dummy text. This is dummy text. This is dummy text.

This  is the    second paragraph. The following sentences are meaningless. The following sentences are meaningless. The following sentences are meaningless.

\end{document}
```

Spaces, Line Breaks, and Paragraphs (Output)

Introduction to LaTeX

Author Name

September 5, 2026

1 Introduction

This is the first paragraph. This is dummy text. This is dummy text. This is dummy text.

This is the second paragraph. The following sentences are meaningless. The following sentences are meaningless. The following sentences are meaningless.

Spaces, Line Breaks, and Paragraphs

- LaTeX treats multiple spaces just like a single space.
- A single line break has the same effect like a single space.
- It doesn't matter how you arrange your text in the editor using spaces or breaks, the output will stay the same.
- A blank line denotes a paragraph break.
- Multiple empty lines are treated as one.

Comments

> Everything following a percent sign (**% symbol** used as comment) until the end of the line will be ignored by LaTeX.

```
\begin{document}
    \maketitle
    \section{Introduction}
    This  is the first paragraph.

    % This  is the    second paragraph.

    This  is the    third paragraph.
\end{document}
```

1 Introduction

This is the first paragraph.
This is the third paragraph.

Writing Special Characters

```
\begin{document}
    \maketitle

    \section{Writing Special Characters}
    Statement \#1: 50\% of \$100 makes \$50. \\
    More special symbols are \&, \_, \{ and \}. \\
    \textbackslash

\end{document}
```

1 Writing Special Characters

Statement #1: 50% of $100 makes $50.
More special symbols are &, _, { and }.
\

Bolds, Italics, and Underlining

```latex
\begin{document}
    \maketitle

    \section{Writing Special Characters}
    Text can be \emph{emphasized}.\\

    \textit{italic} words could be \textbf{bold}.\\
    \textsl{slanted}  or in \textsc{Small Caps}.\\
    Commands can be \textit{\textbf{nested}}.\\
    \emph{Here \emph{emphasizing} nested.}

\end{document}
```

1 Bolds, Italics, and Underlining

Text can be *emphasized.*
italic words could be **bold**.
slanted or in SMALL CAPS.
Commands can be ***nested.***
Here emphasizing *nested.*

Fonts Sizes

> **These commands are “switch command”s**

- applies the command to text following it until the end of the block that the command belongs to

Command	Output
\tiny	Lorem ipsum
\scriptsize	Lorem ipsum
\footnotesize	Lorem ipsum
\small	Lorem ipsum
\normalsize	Lorem ipsum
\large	Lorem ipsum

Command	Output
\Large	Lorem ipsum
\LARGE	Lorem ipsum
\huge	Lorem ipsum
\Huge	Lorem ipsum

Fonts Sizes

> **These commands are "switch command"s**

- applies the command to text following it until the end of the block that the command belongs to

```
\usepackage{blindtext}

\begin{document}

\blindtext[1]

{\tiny This text is tiny.} The effects only
apply to the text in the brackets.

\blindtext[1]

\end{document}
```

it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This text is tiny. The effects only apply to the text in the brackets.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no in-

Fonts Families

typeface = family	command	switch command	output
serif (roman)	`\textrm{Sample Text 0123}`	`\rmfamily`	Sample Text 0123
sans serif	`\textsf{Sample Text 0123}`	`\sffamily`	Sample Text 0123
typewriter (monospace)	`\texttt{Sample Text 0123}`	`\ttfamily`	`Sample Text 0123`

```
\blindtext[1]

This \texttt{function} returns a list of
permutations from 0 to the specified number.

\blindtext[1]
```

It should be written in of the original language. There is no need for special content, but the length of words should match the language.

This `function` returns a list of permutations from 0 to the specified number.

See, this sentence is still in red. Hello, here is some text without a meaning. This text should

Fonts Styles

style	command	switch command	output
medium	`\textmd{Sample Text 0123}`	`\mdseries`	Sample Text 0123
bold	`\textbf{Sample Text 0123}`	`\bfseries`	**Sample Text 0123**
upright	`\textup{Sample Text 0123}`	`\upshape`	Sample Text 0123
italic	`\textit{Sample Text 0123}`	`\itshape`	*Sample Text 0123*
slanted	`\textsl{Sample Text 0123}`	`\slshape`	*Sample Text 0123*
small caps	`\textsc{Sample Text 0123}`	`\scshape`	SAMPLE TEXT 0123

Text Color (switch command)

> `\color{colorname}`: apply the `colorname` to text following the command until the end of the block that the command belongs to.

- `colorname`: specify a named color such as ‘`red`’ or ‘`black`’

```latex
\usepackage{blindtext}
\usepackage{xcolor}

\begin{document}
{\color{blue} This text is in blue.}
But this one is not in blue. However, if you
use the command \color{red} without brackets,
it will be applied to the entire document. \\

See, this sentence is still in red.
\blindtext[1]

\end{document}
```

This text is in blue. But this one is not in blue. However, if you use the command without brackets, it will be applied to the entire document.

See, this sentence is still in red. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Text Color

`\textcolor{colorname}{text}`: sets color of the `text` in the brackets

- `colorname`: specify a named color such as ‘`red`’ or ‘`black`’

```latex
\usepackage{blindtext}
\usepackage{xcolor}

\begin{document}
{\textcolor{blue}{This text is in blue.}
But this one is not blue. However, if you use
the command \textcolor{red}{ without
brackets}, it will be applied to the entire
document. \\

See, this sentence is in black. \blindtext[1]

\end{document}
```

This text is in blue. But this one is not in blue. However, if you use the command without brackets, it will be applied to the entire document.

See, this sentence is in black. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Color Value

- **Default named color (19 colors)**

  `\usepackage{xcolor}`

- **dvipsnames: 68 named colors (CMYK)**

  `\usepackage[dvipsnames] {xcolor}`

- **svgnames: 151 RGB colors**

  `\usepackage[svgnames] {xcolor}`

- **x11names: 317 RGB colors**

  `\usepackage[x11names] {xcolor}`

- **Multiple colors**

  `\usepackage[svgnames, x11names] {xcolor}`

red	yellow	lightgray	pink
green	black	brown	purple
blue	gray	lime	teal
cyan	white	olive	violet
magenta	darkgray	orange	

<Default names>

Apricot	Aquamarine	Bittersweet	Black
Blue	BlueGreen	BlueViolet	BrickRed
Brown	BurntOrange	CadetBlue	CarnationPink
Cerulean	CornflowerBlue	Cyan	Dandelion
DarkOrchid	Emerald	ForestGreen	Fuchsia
Goldenrod	Gray	Green	GreenYellow
JungleGreen	Lavender	LimeGreen	Magenta
Mahogany	Maroon	Melon	MidnightBlue
Mulberry	NavyBlue	OliveGreen	Orange
OrangeRed	Orchid	Peach	Periwinkle
PineGreen	Plum	ProcessBlue	Purple
RawSienna	Red	RedOrange	RedViolet
Rhodamine	RoyalBlue	RoyalPurple	RubineRed
Salmon	SeaGreen	Sepia	SkyBlue
SpringGreen	Tan	TealBlue	Thistle
Turquoise	Violet	VioletRed	White
WildStrawberry	Yellow	YellowGreen	YellowOrange

<dvipsnames names>

Creating Your Own Colors

> **Defining a new color**

`\definecolor{color-name}{color model}{color values}`

```
\usepackage{xcolor}

\definecolor{mypink1}{rgb}{0.858, 0.188, 0.478}
\definecolor{mypink2}{RGB}{219, 48, 122}
\definecolor{mypink3}{cmyk}{0, 0.7808, 0.4429, 0.1412}
\definecolor{mygray}{gray}{0.6}

\begin{document}

User-defined colors with different color models:
\textcolor{mypink1}{Pink with rgb} \\
\textcolor{mypink2}{Pink with RGB} \\
\textcolor{mypink3}{Pink with cmyk} \\
\textcolor{mygray}{Gray with gray} \\

\end{document}
```

color model:

- rgb: three colors in [0, 1]),
- RGB: three colors in [0,255]
- cmyk: four colors in [0, 1]
- gray: grayscale in [0, 1]

User-defined colors with different color models:

1. Pink with rgb
2. Pink with RGB
3. Pink with cmyk
4. Gray with gray

Text Alignments

> **With environment setting:**

- Applied the command between `\begin{}` and `\end{}`

```
\begin{flushleft}
    This    is left-aligned.
\end{flushleft}

\begin{flushright}
    This    is right-aligned.
\end{flushright}
```

```
\begin{center}
    This    is center-aligned.
\end{center}

This is a justified text.
\blindtext[1]
```

This is left-aligned.

This is right-aligned.

This is center-aligned.

This is a justified text. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really?

Text Alignments

> **With switch command**

- Applied the setting to the text following the command until the end of the block that the command belongs to. If there is no block, it lasts until the document ends.

```
\raggedright % “right” for left-aligned
    This    is left-aligned.\\
\raggedleft % “left” for right-aligned
    This    is right-aligned.\\
\centering
    This    is center-aligned.\\
```

```
% return to the default setting
% requires to use ragged2e package
\usepackage{ragged2e}
\justifying
This    is a  justified  text.
```

This is left-aligned.

This is right-aligned.

This is center-aligned.

This is a justified Text. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really?

Content

Greek Letters

αA	`\alpha A`	νN	`\nu N`
βB	`\beta B`	$\xi \Xi$	`\xi \Xi`
$\gamma \Gamma$	`\gamma \Gamma`	$o O$	`o O`
$\delta \Delta$	`\delta \Delta`	$\pi \Pi$	`\pi \Pi`
$\epsilon \varepsilon E$	`\epsilon \varepsilon E`	$\rho \varrho P$	`\rho \varrho P`
ζZ	`\zeta Z`	$\sigma \Sigma$	`\sigma \Sigma`
ηH	`\eta H`	τT	`\tau T`
$\theta \vartheta \Theta$	`\theta \vartheta \Theta`	$\upsilon \Upsilon$	`\upsilon \Upsilon`
ιI	`\iota I`	$\phi \varphi \Phi$	`\phi \varphi \Phi`
κK	`\kappa K`	χX	`\chi X`
$\lambda \Lambda$	`\lambda \Lambda`	$\psi \Psi$	`\psi \Psi`
μM	`\mu M`	$\omega \Omega$	`\omega \Omega`

Arrows

$\leftarrow$	\leftarrow	$\Leftarrow$	\Leftarrow
$\rightarrow$	\rightarrow	$\Rightarrow$	\Rightarrow
$\leftrightarrow$	\leftrightarrow	$\rightleftharpoons$	\rightleftharpoons
$\uparrow$	\uparrow	$\downarrow$	\downarrow
$\Uparrow$	\Uparrow	$\Downarrow$	\Downarrow
$\Leftrightarrow$	\Leftrightarrow	$\Updownarrow$	\Updownarrow
$\mapsto$	\mapsto	$\longmapsto$	\longmapsto
$\nearrow$	\nearrow	$\searrow$	\searrow
$\swarrow$	\swarrow	$\nwarrow$	\nwarrow
$\leftharpoonup$	\leftharpoonup	$\rightharpoonup$	\rightharpoonup
$\leftharpoondown$	\leftharpoondown	$\rightharpoondown$	\rightharpoondown

Miscellaneous symbols

∞	\infty	$\forall$	\forall
$\Re$	\Re	$\Im$	\Im
∇	\nabla	$\exists$	\exists
∂	\partial	$\nexists$	\nexists
$\emptyset$	\emptyset	$\varnothing$	\varnothing
$\wp$	\wp	$\complement$	\complement
$\neg$	\neg	$\cdots$	\cdots
$\square$	\square	$\surd$	\surd
$\blacksquare$	\blacksquare	$\triangle$	\triangle

Binary Operation/ Relation Symbols

$\times$	\times	$\cdot$	\cdot
$\div$	\div	$\cap$	\cap
$\cup$	\cup	$\neq$	\neq
$\leq$	\leq	$\geq$	\geq
$\in$	\in	$\perp$	\perp
$\notin$	\notin	$\subset$	\subset
$\simeq$	\simeq	$\approx$	\approx
$\wedge$	\wedge	$\vee$	\vee
$\oplus$	\oplus	$\otimes$	\otimes
$\Box$	\Box	$\boxtimes$	\boxtimes
$\equiv$	\equiv	$\cong$	\cong

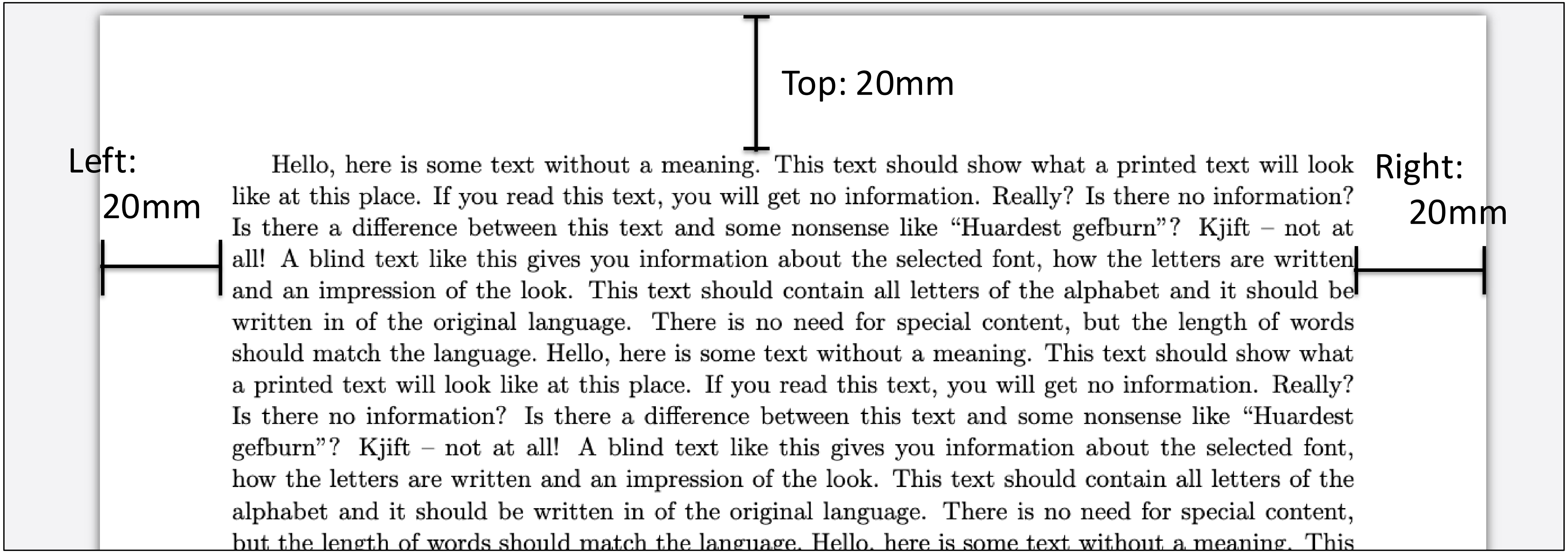
Content

Document Margins

```latex
\documentclass[a4paper, 12pt]{article}

\usepackage{blindtext}
\usepackage{geometry}
\geometry{
    a4paper,
    left=20mm,
    right=20mm,
    top=20mm,
    bottom=20mm
}

\begin{document}
    \blindtext[5]
\end{document}
```

```latex
\documentclass[a4paper,
12pt]{article}
\usepackage[english]{babel}
\usepackage{blindtext}
\usepackage[a4paper,
    left=20mm,
    right=20mm,
    top=20mm,
    bottom=20mm]{geometry}

\begin{document}
    \blindtext[5]
\end{document}
```

Document Margins (Output)



Chapters and Sections

- **\chapter{}**: Applied to book type document
- **\section{}**: First-level section
- **\subsection{}**: Second-level section
- **\subsubsection{}**: Third-level section

Sections in Article

```
\documentclass[a4paper, 12pt]{article}
\usepackage[english]{babel}
\usepackage{blindtext}

\begin{document}

\section{List}
\subsection{Unordered List}  \blindtext[1]
    \subsubsection{Shape}
    \subsubsection{Margin}

\subsection{Ordered List}    \blindtext[1]

\subsection{Detailed List}   \blindtext[1]

\end{document}
```

Sections in Article

1 List

1.1 Unordered List

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

1.1.1 Shape

1.1.2 Margin

1.2 Ordered List

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is

Unordered Section

```
\documentclass[a4paper, 12pt]{article}
\usepackage[english]{babel}
\usepackage{blindtext}

\begin{document}

\section*{List}
\subsection*{Unordered List}  \blindtext[1]
    \subsubsection*{Shape}
    \subsubsection*{Margin}

\subsection*{Ordered List}    \blindtext[1]

\subsection*{Detailed List}   \blindtext[1]

\end{document}
```

Sections in Article

List

Unordered List

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Shape

Margin

Ordered List

Hello, here is some text without a meaning. This text should show what a

Sections in Book

```
\documentclass[a4paper, 12pt]{book}
\usepackage[english]{babel}
\usepackage{blindtext}

\begin{document}

\chapter{Book Chapter}
\section{List}
\subsection{Unordered List}  \blindtext[1]
    \subsubsection{Shape}
    \subsubsection{Margin}
\subsection{Ordered List}    \blindtext[1]
\subsection{Detailed List}   \blindtext[1]

\end{document}
```

Sections in Book

Chapter 1

Book Chapter

1.1 List

1.1.1 Unordered List

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Table of Contents

- Making a table of contents where the command is located

```
\begin{document}

\tableofcontents

\chapter{Book Chapter}
    \section{List}
        \subsection{Unordered List}  \blindtext[1]
            \subsubsection{Shape}
            \subsubsection{Margin}
        \subsection{Ordered List}    \blindtext[1]
        \subsection{Detailed List}   \blindtext[1]
\end{document}
```

Table of Contents (Output)

Contents

Table of Contents

- Unordered sections can be added to Table of Contents

```
\begin{document}

\tableofcontents

\chapter{Book Chapter}
\section{List}
\subsection{Unordered List}  \blindtext[1]
    \subsubsection{Shape}
    \subsubsection{Margin}
\subsection{Ordered List}    \blindtext[1]
\subsection{Detailed List}   \blindtext[1]

\section*{Acknowledgments}
\addcontentsline{toc}{section}{Acknowledgments}

\end{document}
```

Table of Contents (Output)

Contents

Two Columns

> **`\twocolumn`: a page-level two column command**
  - Makes a new page and set two-column page

> **`\onecolumn` makes a new page and set one-column page**

```
\documentclass[a4paper,12pt]{article}
\usepackage{blindtext}

\begin{document}
    \twocolumn
    \blindtext[5]
\end{document}
```

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text

guage. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no

Multiple Columns

> **multicols:** a paragraph-level multi-column command (environmental setting)

```
\documentclass[a4paper,12pt]{article}
\usepackage{blindtext}

\begin{document}
    \section{Multiple Columns Example}
    This is a document with 3 columns.
    \begin{multicols}{3}
        \blindtext[2]
    \end{multicols}

\end{document}
```

Multiple Columns (Output)

9 Multiple Columns Example

This is a document with 3 columns.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there

This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look.

Handling Multiple Files

- **\input{filename}:** imports all the contents from *filename.tex* into the target file
- **\include{filename}:** works like `\input` and put `\clearpage` before and after `\input{filename}`

Handling Multiple Files

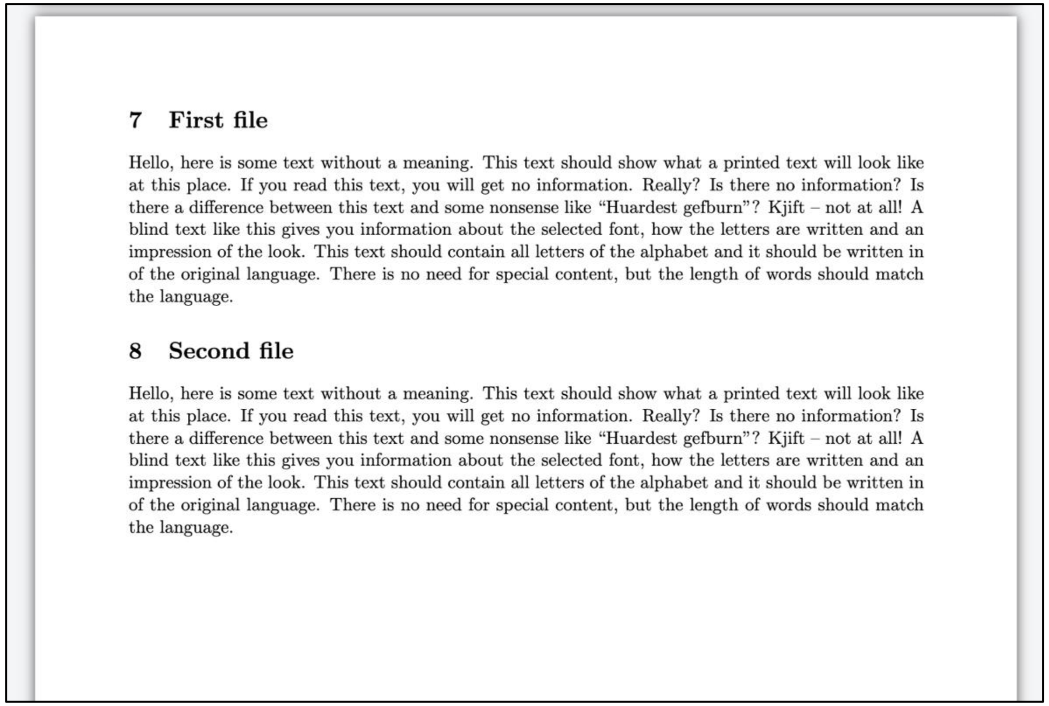
```
\documentclass[a4paper, 12pt]{article}


% Body
\begin{document}
   \maketitle
   \input{section1}
   \input{section2}
   \input{section3}
\end{document}
```

```
\section{Section1}
This section explains
how to write C code.
\blindtext[15]
```

```
\section{Section2}
This section explains
How to write Pseudo-
code.
\blindtext[15]
```

```
\section{Sectin3}
This section explains
how to write Java code.
\blindtext[15]
```

Examples of `\input` and `\include`



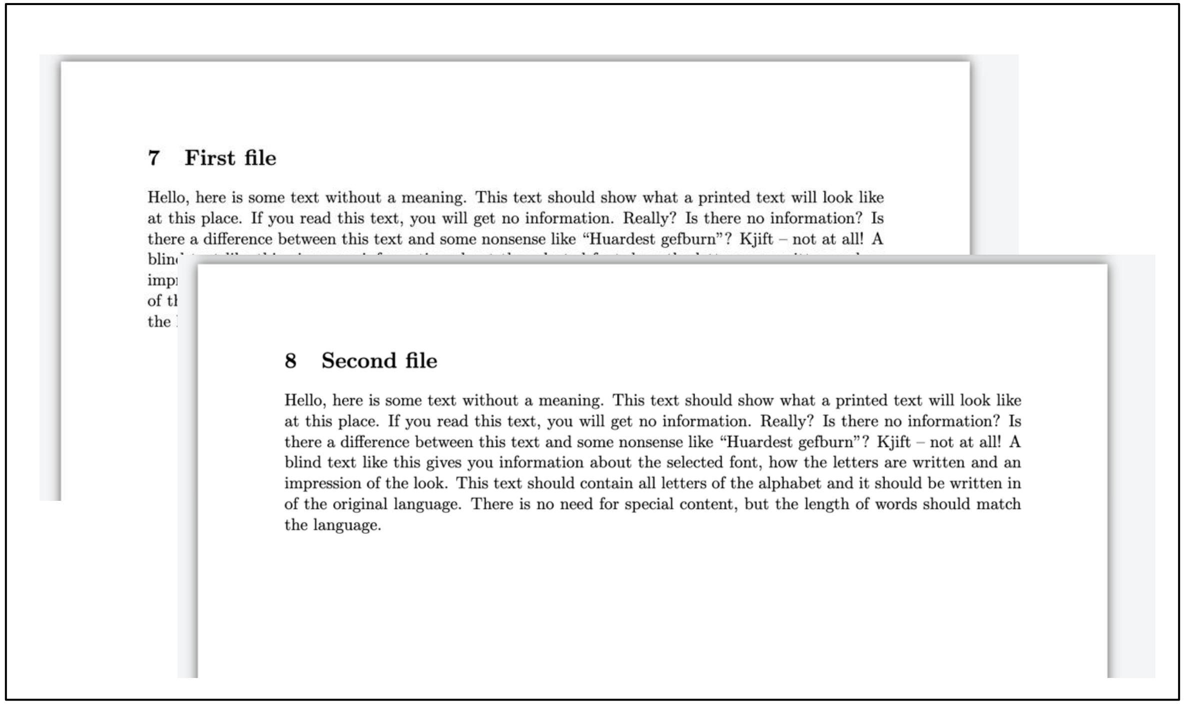
7 First file

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

8 Second file

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

<\input>



7 First file

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blin
impr
of th
the

8 Second file

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

<\include>

Quote Environment

```
…
\begin{quote}
\itshape % switch to italic
Software is prominent in most modern
systems architectures and is often the
primary means for integrating complex
system components. Software engineering and
systems engineering are not merely related
disciplines; they are intimately
intertwined....Good systems engineering is
a key factor in enabling good software
engineering.
\end{quote}
…
```

In modern systems, where concepts such as Edge Computing, Internet of Things and Cyber-physical Systems are prevalent, software is a critical factor. Thus, software engineering is closely related to the Systems Engineering discipline. The Systems Engineering Body of Knowledge claims:

> *Software is prominent in most modern systems architectures and is often the primary means for integrating complex system components. Software engineering and systems engineering are not merely related disciplines; they are intimately intertwined....Good systems engineering is a key factor in enabling good software engineering.*

III. Terminology

A. *Definition*

Notable definitions of software engineering include:

- ”The systematic application of scientific and technological knowledge, methods, and experience to the de-

Content

Custom Commands

```
\newcommand{\TUG}{\textit{TeX   Users  Group}}

\begin{document}
    \section{Creating   Custom Commands}
    The \TUG\ is an organization for people who are interested in \TeX\ or
\LaTeX.
\end{document}
```

- **"\ "**: add space exclusively, it allows line breaking when it is needed.
- **"~"**: add space explicitly, but it does not allow line breaking.

3 Creating Custom Commands

The *TeX Users Group* is an organization for people who are interested in TEX or LATEX.

Creating a Macro

```latex
\newcommand{\keyword}[1]{\textbf{#1}}

\begin{document}

    \section{Defining a Macro and using it}
    \keyword{Grouping} by curly braces limits the
    \keyword{scope} of \keyword{declarations}.

\end{document}
```

4 Defining a Macro and using it

Grouping by curly braces limits the **scope** of **declarations**.

\newcommand

- Defineing a new command
- Only one optional argument is allowed

```
\newcommand{command}[arguments][optional]{definition}
```

`command`	The name of the new command, starting with a backslash followed by lowercase and/or uppercase letters or a backslash followed by a single non-letter symbol. That name must not be already defined and is not allowed to begin with `\end`.
`arguments`	An integer from 1 to 9, the number of arguments of the new command. If omitted, the command will have no arguments.
`optional`	If this is present, then the first of the arguments would be optional with a default value given here. Otherwise all arguments are mandatory.
`definition`	Every occurrence of the `command` will be replaced by `definition` and every occurrence of the form `#n` will then be replaced by the nth argument.

New Command with An Optional Argument

> **The first argument can be optional.**

> **Example: \note command**
- The example has two arguments
- The value of the first argument is determined ‘TODO’ as default

```
\documentclass[a4paper,12pt]{article}
\usepackage{xcolor}

\newcommand{\note}[2][TODO]
{\textbf{\color{red}[#1]} \textit{#2}}

\begin{document}
\note{Need to fix consistency.}
\note[DISCUSS]{This is a command test.}
\end{document}
```

[TODO] *Need to fix consistency.*

[DISCUSS] *This is a command test.*

Content

Lists in LaTeX

- **itemize** environment for creating a bulleted (unordered) list
- **enumerate** environment for creating a numbered (ordered) list
- **description** environment for creating a list of descriptions

Unordered List

```latex
\section{Creating an Unordered List}
Lists are easy to create:
\begin{itemize}
    \item start with the \verb|\item| command.
    \item Entries start with bullets.
    \item The text may be of any length.
\end{itemize}
```

1 Creating an Unordered List

Lists are easy to create:

- start with the `\item` command.
- Entries start with bullets.
- The text may be of any length.

Ordered List

```latex
\section{Creating an Ordered List}
Numbered (ordered) lists are easy to create:
\begin{enumerate}
    \item Items are numbered automatically.
    \item start at 1 with each \texttt{enumerate}.
    \item Another entry in the list
\end{enumerate}
```

2 Creating an Ordered List

Numbered (ordered) lists are easy to create:

1. Items are numbered automatically.
2. start at 1 with each `enumerate`.
3. Another entry in the list

Description List

```
\section{Description List}
This is an entry \textit{without} a label.
\begin{description}
    \item[Something short] A short one-line description.
    \item[Something long] A much longer description.
    \blindtext[1]
\end{description}
```

3 Description List

This is an entry *without* a label.

Something short A short one-line description.

Something long A much longer description. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information

Content

Inserting Images

> **Need to use `graphicx` package**

- Available to specify the default path for images (current directory without the setting)

> **`\includegraphics{[path/]name}`**

- It finds an image named `name` in the default path (no need for file extensions)
- `path` is optional. The default path will be the root.

```
\usepackage{graphicx}
\graphicspath{ {./images/} }   % set default path for images

\begin{document}
\section{Inserting Images in a document}
The  universe is immense and it seems  to be homogeneous, in a  large scale, everywhere  we
look at.
\includegraphics{galaxy}    % put an image from “./images/galaxy.*”
There's a picture of a galaxy above
\end{document}
```

Inserting Images (Output)

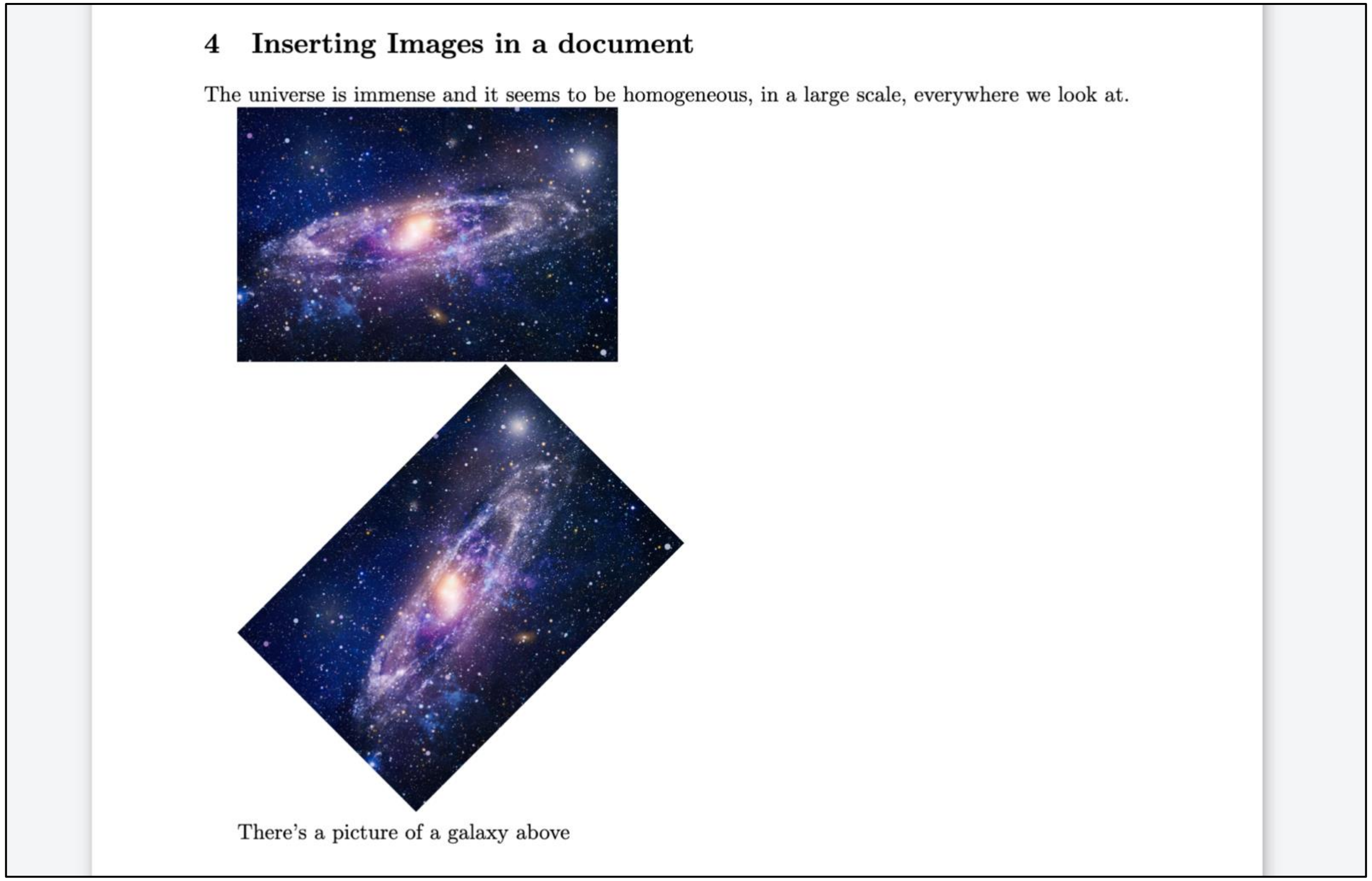
4 Inserting Images in a document

The universe is immense and it seems to be homogeneous, in a large scale, everywhere we look at.

There's a picture of a galaxy above

Changing the Image Size and Rotation

```
\includegraphics[width=0.8\linewidth]{galaxy}
\includegraphics[scale=0.5, angle=45]{galaxy}
```



4 Inserting Images in a document

The universe is immense and it seems to be homogeneous, in a large scale, everywhere we look at.

There's a picture of a galaxy above

Figure Environment

> **Figures are visual presentations**

- Images, chart, pseudo-code, and etc.

> **Figure Environment usually comes with caption, label, location, and etc.**

- `\caption{text}`: set `text` as a caption
- `\ref{label}`: put the number of the specified label

```
\begin{figure}[h]
    \centering
    \includegraphics[width=0.3\linewidth]{galaxy}
    \caption{A nice plot.}
    \label{fig:mesh1}
\end{figure}
The following Figure~\ref{fig:mesh1} is a nice photo. \blindtext[5]
```

Figure Environment

6 Location of figures

The following Figure 1 is a nice photo.

Figure 1: A nice plot.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be

Positioning Figures

Parameter	Position
h	Place the float *here*, i.e., *approximately* at the same point it occurs in the source text (however, not *exactly* at the spot)
`t`	Position at the *top* of the page.
b	Position at the *bottom* of the page.
p	Put on a special *page* for floats only.
`!`	Override internal parameters LaTeX uses for determining "good" float positions.
`H`	Places the float at precisely the location in the $\LaTeX$ code. Requires the `float` package. This is somewhat equivalent to `h!`.

Positioning Figures in Multiple Columns

> **With two columns, the figure located within a column**

- Note that "t" is assigned for the figure loaction

```
\twocolumn

\section{Location of figures}
The following Fig~\ref{fig:galaxy} is a nice photo.
\begin{figure}[t]
    \centering
    \includegraphics[width=0.6\linewidth]{galaxy}
    \caption{A nice plot.}
    \label{fig:galaxy}
\end{figure}
\blindtext[15]
```

Positioning Figures in Multiple Columns

Figure 1: A nice plot.

6 Location of figures

The following Figure 1 is a nice photo.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no in-

of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like

Positioning Figures in Multiple Columns

- "*" is a special version of an environmental command, which makes it span across all columns in a multi-column layout.

```
\twocolumn

\section{Location of figures}
The following Fig~\ref{fig:galaxy} is a nice photo.
\begin{figure*}[t]
    \centering
    \includegraphics[width=0.4\linewidth]{galaxy}
    \caption{A nice plot.}
    \label{fig:galaxy}
\end{figure*}
\blindtext[15]
```

Positioning Figures in Multiple Columns

Figure 1: A nice plot.

A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words

the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some

Sub Figures

- **Use package subcaption**

  ```
  \usepackage{subcaption}
  ```

- **subfigure environment can be used in a figure environment**

- **Additional commands can be used to control locations**
  - **\hspace{size}**: add or remove space between objects (horizontal)
  - **\vspace{size}**: add or remove space between objects (vertical)
  - **\hfill**: fill all the remaining space between objects (horizontal)
  - **\vfill**: fill all the remaining space between objects (vertical)

Sub Figures

```
\usepackage{subcaption}
\begin{figure*}[h!]
    \begin{center} % center environment for subfigures
        \begin{subfigure}[t]{\columnwidth} % new subfigure starts
            \centering
            \includegraphics[width=0.88\columnwidth]{galaxy.jpg}
            \caption{Galaxy 1}
            \label{fig:galaxy2 type1} % label for subfigure
        \end{subfigure}
        \hfill     % fill up the space between subfigures
        \begin{subfigure}[t]{\columnwidth} % new subfigure starts
            \centering
            \includegraphics[width=0.72\columnwidth]{NGC}
            \caption{galaxy2 2} % caption for subfigure
            \label{fig:galaxy type2} % label for subfigure
        \end{subfigure}
    \end{center}
    \caption{Two galaxies} % caption of the entire figure
    \label{fig:galaxy2}    % this label can be referred in text
\end{figure*}
```

Sub Figures

(a) Galaxy 1

(b) galaxy2 2

Figure 2: Two galaxies

at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all!

guage. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like

Content

`tabular` Environment

> **A default method to create tables**

> **The argument**

- Specifying the number of columns by using alignment symbols
- Alignment symbols:
  - `l` (left), `r` (right), `c` (center)
- '|': inserts a vertical line (e.g., {|c|c|c|})

> **Commands for cells / rows**

- &: cell separator
- \hline: inserts a horizontal line
- \\ : row separator

```
\begin{tabular}{ccc}
    \hline
    title1 & title2 & title3 \\
    \hline
    cell1 & cell2 & cell3 \\
    cell4 & cell5 & cell6 \\
    cell7 & cell8 & cell9 \\
    \hline
\end{tabular}
```

title1	title2	title3
cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

tabular Environment

> **Example of vertial lines**

```
\begin{tabular}{|c|c|c|}
    \hline
    title1 & title2 & title3 \\
    \hline
    cell1 & cell2 & cell3 \\
    cell4 & cell5 & cell6 \\
    cell7 & cell8 & cell9 \\
    \hline
\end{tabular}
```

title1	title2	title3
cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Utilizing booktabs

> **Used for ACM/IEEE journal papers**

> **Providing high-quality table commands**

- Fancy horizontal lines
- Recommands not to use vertical lines

> **Meaningful horizontal lines**

- `\toprule`: top line of a table
- `\midrule`: separete title and body
- `\bottomrule`: bottom line of a table
- `\cmidrule{a-b}`: lines across a-b cells

```
\usepackage{booktabs}

\begin{tabular}{ccc}
    \toprule
    title1 & title2 & title3 \\
    \midrule
    cell1 & cell2 & cell3 \\
    cell4 & cell5 & cell6 \\
    cell7 & cell8 & cell9 \\
    \bottomrule
\end{tabular}
```

title1	title2	title3
cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Merging Columns

\multicolumn{count}{alignment}{text}

- ***count***: number of columns to merge
- ***alignment***: center(c), left(l), or right(r)
- ***text***: text to be appeared in the merged column

title group		
title1	title2	title3
cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

```
\usepackage{booktabs}

\begin{tabular}{c|c|c}
    \toprule
    \multicolumn{3}{c}{title group}\\
    \cmidrule{1-3}
    title1 & title2 & title3 \\
    \midrule
    cell1 & cell2 & cell3 \\
    cell4 & cell5 & cell6 \\
    cell7 & cell8 & cell9 \\
    \bottomrule
\end{tabular}
```

Merging Rows

\multirow{count}{width}{text}

- ***count***: number of rows to merge
- ***width***: size of width
- ***text***: text to be appeared in the merged row

```
\usepackage{booktabs}
\usepackage{multirow}

\begin{tabular}{c|c|c}
    \toprule
    title1 & title2 & title3 \\
    \midrule
    \multirow{3}{4em}{Merged row}
       & cell2 & cell3 \\
       & cell5 & cell6 \\
       & cell8 & cell9 \\
    \bottomrule
\end{tabular}
```

The merged cells should be kept empty

title1	title2	title3
Merged row	cell2	cell3
	cell5	cell6
	cell8	cell9

table Environment

- ***caption{}***: title of the table
- ***label{}***: label for referencing
- Positioning options are the same as in the figure environment
- The numbering of tables is independent of figures, equations, and other elements.

```latex
\begin{table}[t!]
\centering
\caption{Example Table}
\label{tab:table1}
\begin{tabular}{ccc}
    \toprule
    \multicolumn{3}{c}{group} \\
    \cmidrule{1-3}
    title1 & title2 & title3 \\
    \midrule
    cell1 & cell2 & cell3 \\
    cell4 & cell5 & cell6 \\
    cell7 & cell8 & cell9 \\
    \bottomrule
\end{tabular}
\end{table}
Table~\ref{tab:table1} is an example.
```

Table Environment (Output)

Table 1: Example Table

title group		
title1	title2	title3
cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Table 1 is an example. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference be-

meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no

Content

Mathematical Modes

> **LATEX allows two writing modes for mathematical expressions:**

- ***inline*** math mode is used to write formulas that are part of a paragraph
- ***display*** math mode is used to write expressions that are not part of a paragraph, and are therefore put on separate lines

Inline Math Mode

You can use any of these "delimiters" to typeset your math in inline mode:

- `\(...\)`
- `$...$`
- `\begin{math}...\end{math}`

```
There are three types of commands for printing mathematical
equations:
\begin{math} x^2+y^2=1 \end{math},
\( x^2+y^2=1 \),
and $ x^2+y^2=1 $.
These three commands have the same effect, but different
notation. LaTeX provides several equivalent notations for
entering inline math mode, partly for historical compatibility
with TeX and for convenient syntax. Users prefer to use simpler
ones: \( \) and $  $.
```

There are three types of commands for printing mathematical equations: $x^2 + y^2 = 1$, $x^2 + y^2 = 1$, and $x^2 + y^2 = 1$. These three commands have the same effect, but different notation. LaTeX provides several equivalent notations for entering inline math mode, partly for historical compatibility with TeX and for convenient syntax. Users prefer to use simpler ones: and .

Display Math Mode

- **Use one of these constructions:**
  - Unnumbered equations

```
\[...\]
$$ ... $$
\begin{displaymath}...\end{displaymath}
```

  - Numbered equations

```
\begin{equation}...\end{equation}
```

Mathematical Expressions

```latex
\begin{document}

    The Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents.
    Meaning the next equation has no integer solutions:

    \[ x^n + y^n = z^n \]

\end{document}
```

The Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n$$

Display Math Mode (with equation number)

```
The mass-energy equivalence is described by the equation
\begin{equation}
    E=mc^2
\end{equation}
discovered in 1905 by Albert Einstein.
\begin{equation}
    E=m
\end{equation}
```

The mass-energy equivalence is described by the equation

$$E = mc^2 \tag{1}$$

discovered in 1905 by Albert Einstein.

$$E = m \tag{2}$$

Subscript and Superscript

> **Subscript(_)**
- Indicates the starting point of integral or sum
- Indicates a base value of log and others

> **Superscript(^)**
- Indicates the ending point of integral or sum
- Indicates exponent

> **{}**: grouping equations

> **\sum**: sum notation (summation over a range)

> **\int**: integral notation (continuous accumulation over an interval)

```
\[ y=\sum_0^{10}   { x^2 + y^2 } \]
\[ y=\int_0^1   { x^2 + y^2 \: dx } \]
```

$$y = \sum_0^{10} x^2 + y^2$$

$$y = \int_0^1 x^2 + y^2 \: dx$$

Fractions and Other Commands

- `\frac{numerator}{denominator}`
- `\infty`: infinity
- `\prod`: product notation (multiplication over a range)

$$\sum_{i=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$$

$$\frac{1+\frac{a}{b}}{1+\frac{1}{1+\frac{1}{a}}}$$

```
\begin{displaymath}
    \sum_{i=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1- p^{-s}}
\end{displaymath}

\[
     \frac{ 1+\frac{a}{b}  }{ 1+\frac{1}{ 1+\frac{1}{a} }  }
\]
```

Brackets and Parentheses

<table>
<tr><th>Type</th><th>LaTeX markup</th><th>Renders as</th></tr>
<tr><td>Parentheses; round brackets</td><td><code>(x+y)</code></td><td>$(x+y)$</td></tr>
<tr><td>Brackets; square brackets</td><td><code>[x+y]</code></td><td>$[x+y]$</td></tr>
<tr><td>Braces; curly brackets</td><td><code>\{ x+y \}</code></td><td>$\{x+y\}$</td></tr>
<tr><td>Angle brackets</td><td><code>\langle x+y \rangle</code></td><td>$\langle x+y \rangle$</td></tr>
<tr><td>Pipes; vertical bars</td><td><code>|x+y|</code></td><td>$\lvert x+y \rvert$</td></tr>
<tr><td>Double pipes</td><td><code>\|x+y\|</code></td><td>$\lVert x+y \rVert$</td></tr>
</table>

Brackets and Parentheses (cont')

> **Adjustable size of brackets**

```
\[
    F   =  G \left(\frac{m_1  m_2}{r^2} \right)
\]

\[
    \left[\frac{ N } {\left(\frac{L}{p} \right)  - (m+n) } \right]
\]
```

$$F = G \left(\frac{m_1 m_2}{r^2} \right)$$

$$\left[\frac{ N } {\left(\frac{L}{p} \right) - (m+n) } \right]$$

Additional Size of Brackets and Parentheses

<table>
<tr><th>LaTeX markup</th><th>Renders as</th></tr>
<tr><td><code>\bigl(\Bigl(\biggl(\Biggl(</code></td><td>$\bigl(\Bigl(\biggl(\Biggl($</td></tr>
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<tr><th>LaTeX markup</th><th>Renders as</th></tr>
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</table>

Aligning Equations with amsmath package

> **When writing multi-line equations, we may need `align` environment**

> The & tells LaTeX **where to align** equations vertically.
>
> - Left side of & aligns to right
> - Right side of & aligns to left

> **The numbering is shared with equation environment**

> **Each line is assigned new number.**

```
\usepackage{amsmath}
\begin{align}
    x&=y          \\
    2x&=-y        \\
    -4 + 5x&=2+y
\end{align}
```

$$
\begin{align}
x &= y \tag{5}\\
2x &= -y \tag{6}\\
-4 + 5x &= 2 + y \tag{7}
\end{align}
$$

Aligning Equations with amsmath package

> **The `align*` environment shows equations without numbering**

> **Multiple & can be used for multi-column alignment**

- The first & makes a pair with left and right items
- The second & splits columns

```
\usepackage{amsmath}
\begin{align*}
    x&=y            & w &=z               & a&=b+c   \\
    2x&=-y          & 3w&=\frac{1}{2}z    & a&=b     \\
    -4 + 5x&=2+y    & w+2&=-1+w           & ab&=cb
\end{align*}
```

$$\begin{aligned} x &= y & w &= z & a &= b + c \\ 2x &= -y & 3w &= \frac{1}{2}z & a &= b \\ -4 + 5x &= 2 + y & w + 2 &= -1 + w & ab &= cb \end{aligned}$$

Spacing Commands (use amsmath package)

LaTeX code	Description
`\quad`	space equal to the current font size (= 18 mu)
`\,`	3/18 of `\quad` (= 3 mu)
`\:`	4/18 of `\quad` (= 4 mu)
`\;`	5/18 of `\quad` (= 5 mu)
`\!`	-3/18 of `\quad` (= -3 mu)
`\ ` (space after backslash!)	equivalent of space in normal text
`\qquad`	twice of `\quad` (= 36 mu)

```
\[
   y=\int_0^1 {x^2 + y^2 \: dx}
\]
```

$$y = \int_0^1 x^2 + y^2 \: dx$$

Spacing Commands

```latex
\begin{align*}
    f(x) &= x^2\! +3x\! +2 \\
    f(x) &= x^2+3x+2 \\
    f(x) &= x^2\, +3x\, +2 \\
    f(x) &= x^2\: +3x\: +2 \\
    f(x) &= x^2\; +3x\; +2 \\
    f(x) &= x^2\ +3x\ +2 \\
    f(x) &= x^2\quad +3x\quad +2 \\
    f(x) &= x^2\qquad +3x\qquad +2
\end{align*}
```

$$
\begin{aligned}
f(x) &= x^2\! +3x\! +2 \\
f(x) &= x^2+3x+2 \\
f(x) &= x^2\, +3x\, +2 \\
f(x) &= x^2\: +3x\: +2 \\
f(x) &= x^2\; +3x\; +2 \\
f(x) &= x^2\ +3x\ +2 \\
f(x) &= x^2\quad +3x\quad +2 \\
f(x) &= x^2\qquad +3x\qquad +2
\end{aligned}
$$

Invisible Brackets (\left. and \right.)

> **Each line of multi-line equations is treated as an individual equation**

- Missing brackets causes errors (Brackets must appear in pairs)

> **Invisible brackets (`\left.` and `\right.`) ensure proper pairing without displaying brackets**

```
\usepackage{amsmath}

\begin{align*}
    y = 1 + & \left( \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \ldots \right. \\
            & \left. \quad + \frac{1}{x^{n-1}} + \frac{1}{x^n} \right)
\end{align*}
```

$$
\begin{aligned}
y = 1 + & \left(\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \ldots \right. \\
& \left. \quad + \frac{1}{x^{n-1}} + \frac{1}{x^n} \right)
\end{aligned}
$$

amsmath matrix environments

> **Need to be located in `equation` environment**

Type	$\LaTeX$ markup	Renders as

Plain	`\begin{matrix}`
	`1 & 2 & 3\\`
	`a & b & c`
	`\end{matrix}`
		$\begin{matrix} 1 & 2 & 3\\ a & b & c \end{matrix}$

Parentheses; round brackets	`\begin{pmatrix}`
	`1 & 2 & 3\\`
	`a & b & c`
	`\end{pmatrix}`
		$\begin{pmatrix} 1 & 2 & 3\\ a & b & c \end{pmatrix}$

Brackets; square brackets	`\begin{bmatrix}`
	`1 & 2 & 3\\`
	`a & b & c`
	`\end{bmatrix}`
		$\begin{bmatrix} 1 & 2 & 3\\ a & b & c \end{bmatrix}$

amsmath matrix environments

> **Need to be located in `equation` environment**

Type	LaTeX markup	Renders as
Braces; curly brackets	`\begin{Bmatrix}` `1 & 2 & 3\\` `a & b & c` `\end{Bmatrix}`	$\begin{Bmatrix} 1 & 2 & 3\\ a & b & c \end{Bmatrix}$
Pipes	`\begin{vmatrix}` `1 & 2 & 3\\` `a & b & c` `\end{vmatrix}`	$\begin{vmatrix} 1 & 2 & 3\\ a & b & c \end{vmatrix}$
Double pipes	`\begin{Vmatrix}` `1 & 2 & 3\\` `a & b & c` `\end{Vmatrix}`	$\begin{Vmatrix} 1 & 2 & 3\\ a & b & c \end{Vmatrix}$

amsmath matrix environments

- Need to be located in `equation` environment

```
\usepackage{amsmath}

\[
A=
\begin{pmatrix}
  1 & 2 & 3\\
  a & b & c
\end{pmatrix}
\]
```

$$A = \begin{pmatrix} 1 & 2 & 3 \\ a & b & c \end{pmatrix}$$

Content

listings

A built-in LaTeX package for typesetting source code. It provides basic syntax highlighting (keywords/comments/strings), line numbers, frames, and file-import. Additionally, users can customize the style conveniently.

> `\lstlisting` environment

- language: C, Python, Java, and so on.
- caption: caption for the code
- label: label for the code

```
\usepackage{listing}

\begin{lstlisting}[
    language=C,
    caption={Simple sum},
    style=CStyle,
    label={lst:sum}
]
#include <stdio.h>
int main(void){
    int a=10, b=20;
    printf("%d\n", a+b);
    return 0;
}
\end{lstlisting}
```

Listing 1: Simple sum

```
#include <stdio.h>
int main(void){
    int a=10, b=20;
    printf("%d\n", a+b);
    return 0;
}
```

listings

A built-in LaTeX package for typesetting source code. It provides basic syntax highlighting (keywords/comments/strings), line numbers, frames, and file-import. Additionally, users can customize the style conveniently.

> `\lstinputlisting` command
  - firstline: the first line number to show
  - lastline: the lset line number to show

Listing 1: Simple sum

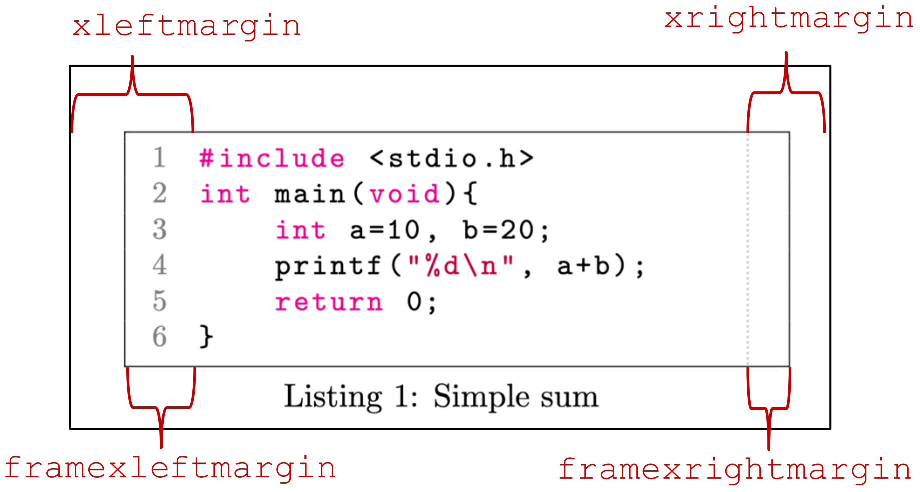
```c
#include <stdio.h>
int main(void){
    int a=10, b=20;
    printf("%d\n", a+b);
    return 0;
}
```

```latex
\usepackage{listing}

% load code from the selected
lines of file “code/main.c”
\lstinputlisting[
    language=C,
    caption={Simple sum},
    label={lst:sum},
    firstline=1,
    lastline=10
]{code/main.c}
```

Customizing listing styles

```
\lstdefinestyle{CStyle}{                    % Set style name------
  language=C,                               % default target language
  backgroundcolor=\color{white},            % background color
  commentstyle=\color{teal},                % color for Comments
  keywordstyle=\color{magenta},             % color for Keywords
  stringstyle=\color{purple},               % color for strings
  basicstyle=\small\ttfamily,               % default font
  breaklines=true,                          % break a long line
  breakatwhitespace=false,                  % line break only at space
  keepspaces=true,                          % keep continuous spaces
  numberstyle=\small\color{gray},           % color for line numbers
  numbers=left,                             % location of line numbers
  numbersep=1em,                            % margin with line numbers
  showspaces=false,                         % show a symbol for space in code
  showstringspaces=false,                   % show a symbol for space in string
  showtabs=false,                           % show a symbol for tabs
  tabsize=4,                                % number of spaces for a tab
  xleftmargin=2.5em,                        % left margin between the listing object and code
  xrightmargin=0.5em,                       % right margin between the listing object and code
  framexleftmargin=2em,                     % left margin between border and code
  frame=tb,                                 % draws borders (t: top, b: bottom, l: left, r:right)
  linewidth=\textwidth                      % width of the listing object
  captionpos=b,                             % location of the caption
}
```



Customizing listing styles

> Setting default style

```
% Set default style
\lstset{style=CStyle}
```

> Setting style with option

```
\usepackage{listing}

% load code from the selected lines
of file “code/main.c”
\lstinputlisting[
    language=C,
    caption={Simple sum},
    label={lst:sum},
    style=CSstyle,  % set style
    firstline=1,
    lastline=10
]{code/main.c}
```

```
#include <stdio.h>
int main(void){
    int a=10, b=20; //int
    printf("%d\n", a+b);
    return 0;
}
```

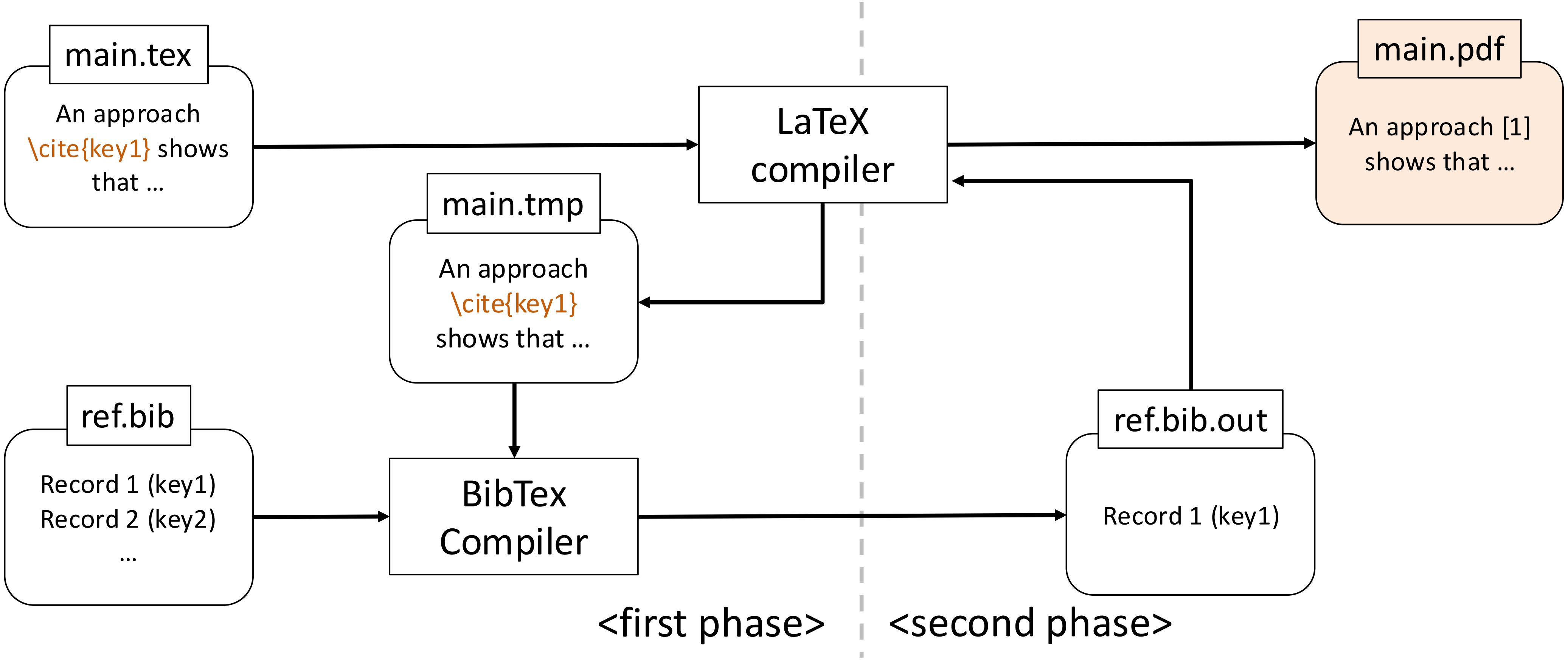
Listing 1: Simple sum

Content

Bibliographies in LaTeX

> LaTeX uses the tool BibTeX for managing bibliographic references

> Basic workflow



Making bib Files

> Bibliography files are organized in blocks, with different fields depending on reference type

- Book (@book), Journal paper (@article)

```
@book{ammann:2016:introduction,     % key value
  title     = {Introduction to software testing},
  author    = {Ammann, Paul and Offutt, Jeff},
  year      = {2016},
  publisher = {Cambridge University Press}
}
```

```
@article{mateo:2012:validating,     % key value
    title       = {Validating second-order mutation at system level},
    author      = {Mateo, Pedro Reales and Usaola, Macario Polo and Aleman,
Jose Luis Fernandez},
    journal     = {IEEE Transactions on Software Engineering},
    volume      = {39},
    number      = {4},
    pages       = {570--587},
    year        = {2012},
    publisher   = {IEEE}
}
```

Making bib Files (cont’)

> Bibliography files are organized in blocks, with different fields depending on reference type

- Conference paper (@inproceedings)

```
@inproceedings {jaekwon:2023:MOTIF,
    author    = {Jaekwon Lee and Enrico Vigano and Oscar Cornejo and Fabrizio Pastore and Lionel
Briand},
    booktitle = {2023 38th IEEE/ACM International Conference on Automated Software Engineering (ASE)},
    title     = {Fuzzing for CPS Mutation Testing},
    year      = {2023},
    volume    = {},
    issn      = {},
    pages     = {1377-1389},
    doi       = {10.1109/ASE56229.2023.00079},
    url       = {https://doi.ieeecomputersociety.org/10.1109/ASE56229.2023.00079},
    publisher = {IEEE Computer Society},
    address   = {Los Alamitos, CA, USA},
    month     = {sep}
}
```

Making bib Files (cont')

> Bibliography files are organized in blocks, with different fields depending on reference type

- Thesis (@phdthesis), Others (@mics)

```
@phdthesis{Ficici:2004:thesis,
    author   = {Ficici, Sevan Gregory},
    school   = {Brandeis University, Department of
Computer Science, Waltham, MA},
    title    = {Solution Concepts in Coevolutionary
Algorithms},
    type     = {Ph.D. thesis},
    year     = {2004}
}
```

```
mics{web:lang:stats,
    author = {W3Techs},
    title  = {Usage Statistics of Content
Languages for Websites},
    year   = {2017},
    note   = {Last accessed 16 September 2024},
    url    =
{http://w3techs.com/technologies/overview/content
_language/all}
}
```

Using bib in LaTeX

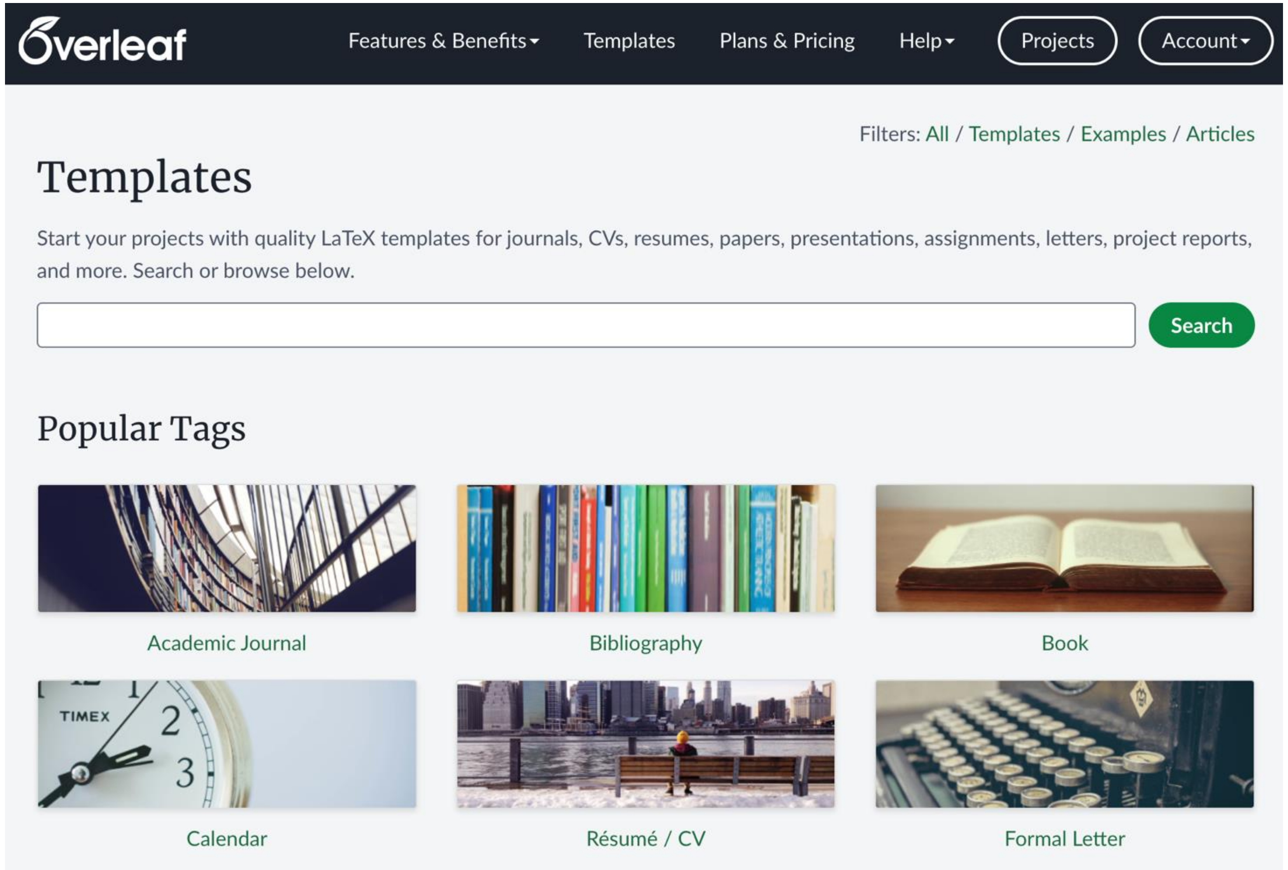
> **There are many bibliography styles to choose from:**
  - Common LaTeX options include plain “[42]” and APA like “[Author, 2019]”
  - Many publishers provide their own styles that should be used

```
Citations are used via their keys \cite{keyValue}.
Never use multiple cite commands to several references in one place.
Instead, follow the way of \cite{keyValue1, keyValue2, …, keyValueN}
…
% At the place where the literature list should appear
\bibliographystyle{plain} % selects citation style
\bibliography{references} % loads files references.bib
```

Content

Selecting a Template

Class (Template) provides predefined design of documents



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Abstract This demo file is intended to serve as a "starter file" for IEEE conference papers produced under LaTeX.

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Tags Conference Paper | IEEE Official Templates | IEEE (all)

Conference Paper Title*

*Note: Sub-titles are not captured in Xplore and should not be used

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name of organization (of Aff.)
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name of organization (of Aff.)
City, Country
email address or ORCID

6th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

This document is a model and instructions for LaTeX. Please observe the conference page limits.

II. Ease of Use

A. Maintaining the Integrity of the Specifications

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

III. Prepare Your Paper Before Styling

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads—LaTeX will do that for you.

Identify applicable funding agency here. If none, delete this.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

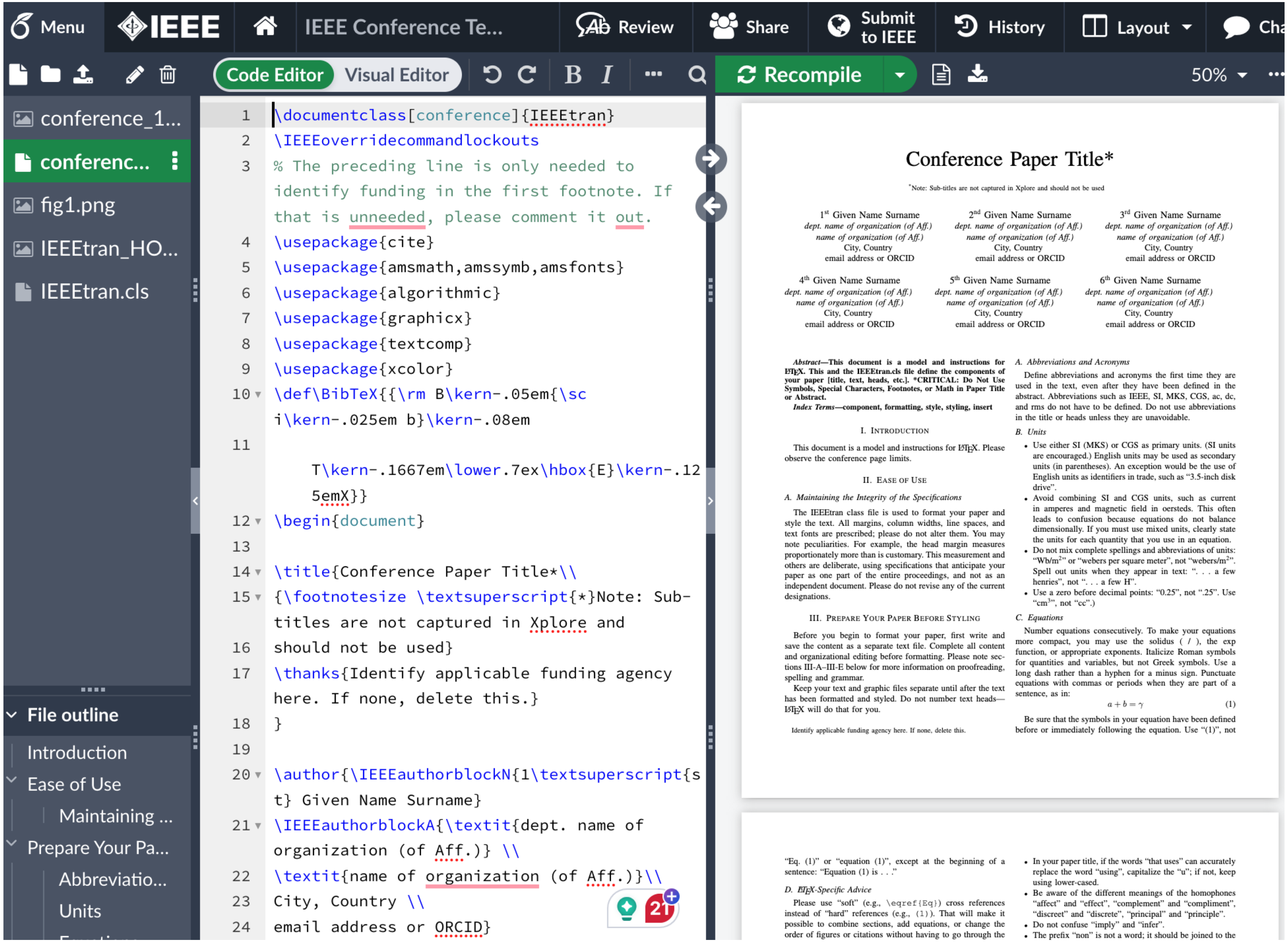
- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: "Wb/m^2" or "webers per square meter", not "$webers/m^2$". Spell out units when they appear in text: ". . . a few henries", not ". . . a few H".
- Use a zero before decimal points: "0.25", not ".25". Use "cm^3", not "cc".)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (1)$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(1)", not



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Abbreviatio...
Units
\documentclass[conference]{IEEEtran}
\IEEEoverridecommandlockouts
% The preceding line is only needed to identify funding in the first footnote. If that is unneeded, please comment it out.
\usepackage{cite}
\usepackage{amsmath,amssymb,amsfonts}
\usepackage{algorithmic}
\usepackage{graphicx}
\usepackage{textcomp}
\usepackage{xcolor}
\def\BibTeX{{\rm B\kern-.05em{\sc i\kern-.025em b}\kern-.08em
T\kern-.1667em\lower.7ex\hbox{E}\kern-.125emX}}
\begin{document}
\title{Conference Paper Title*\\
{\footnotesize *Note: Sub-titles are not captured in Xplore and should not be used}
\thanks{Identify applicable funding agency here. If none, delete this.}
}
\author{\IEEEauthorblockN{1st Given Name Surname}
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\textit{name of organization (of Aff.)}\\
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Abstract—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.
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I. Introduction
II. Ease of Use
A. Maintaining the Integrity of the Specifications
III. Prepare Your Paper Before Styling
A. Abbreviations and Acronyms
B. Units
C. Equations
D. LaTeX-Specific Advice

Summary

> LaTeX can be used to typeset tables and figures

> Custom commands and includes simplify work on LaTeX projects

> LaTeX excels at typesetting mathematical expressions

> BibTeX is used to manage citations in LaTeX

Practices

> Create a **1-page team introduction poster**

> **Create a document that explains a concept**
- Please pick a Wiki page that includes equations, tables, and figures.
- Reproduce the selected document using LaTeX
  - Include at least three references
  - Include at least four sections
  - Add some sentences that refer to figures and tables. (highlights with colors)
  - DO NOT NEED TO reproduce all the sections
  - Example: https://en.wikipedia.org/wiki/Factorial

> **Submit each result at the corresponding slot on the eRuri website.**

QUESTIONS & ANSWERS